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Photodynamic antimicrobial chemotherapy (PACT) using riboflavin inhibits the mono and dual species biofilm produced by antibiotic resistant *Staphylococcus aureus* and *Escherichia coli*

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ABSTRACT

Purpose: Multispecies biofilms play a significant role in persistent infections. Furthermore, by interspecies transfer of antibiotic resistance genes, multispecies biofilms spread antibiotic resistance. The purpose of this study was to investigate the effect of Photodynamic Antimicrobial Chemotherapy (PACT) using riboflavin on mono and multi species biofilms.
Methods: For this we used two clinically relevant opportunistic pathogens species *E. coli* and *S. aureus* as mono-species and multispecies biofilms. We did broth dilution assay for antibacterial, crystal violet assay for biofilms and fluorometric study for reactive oxygen species (ROS) and extracellular polymeric substance (EPS) production by phenol−HCl method.
Results: Antibacterial study revealed that photo-illuminated riboflavin shows bactericidal effect against each bacteria and their mix culture. *E. coli* was found to be little more resistant than *S. aureus*. Crystal violet assay revealed photo-illuminated riboflavin shows anti-biofilms activity against both mono and mix species biofilms. But mix species biofilms were more resistant to PACT than mono species biofilms. Further study revealed this may be due to the interaction between different EPS production, hence in mix species biofilms EPS production is less affected after PACT than mono species biofilms. We found photo-illuminated riboflavin increased the intracellular ROS production.
Conclusion: Photo-illuminated riboflavin shows bactericidal and anti-biofilms effect against each bacteria and their mix culture. Photo-illuminated increased intracellular ROS production, which may induce the oxidative stress and destroy the respiratory system of bacteria

1. Introduction

Biofilm is a three-dimensional and heteromorphic communities of bacteria embedded in a matrix made from self-producing extracellular polymeric substance (EPS) [1]. Due to the expression of specific genes, the sessile form of bacteria within the biofilms show different characteristics from those under planktonic conditions [2]. It has been reported that in comparison with organisms in suspension, the bacteria associated with the biofilms exhibit much more resistance to several antimicrobial agents, different physical factors, deprivation of nutrients, changes in pH, reactive oxygen radicals and the host immune system [1,3]. Moreover, 65 % of infections are connected with biofilms, and that often cause serious illness with an extended hospital stay, increased costs, and high mortality rate [4,5].

Previously, as because of the extensive use of culture-dependent isolation techniques, nearly all infectious diseases have been recognised as mono microbial in nature. But, the concept of poly microbial nature of the infectious diseases are emerge and increasing recently, due to the recent advancement of culture-independent analysis [6]. Poly microbial biofilms is a mixed microbial species community and may considered as the most dominant form of microbial life in nature [7]. Poly microbial biofilms have several benefits, for instance it has enlarged gene pool with more efficient DNA sharing, quorum sensing, more resistant to the antibacterial agents, metabolic cooperation, and several other synergies [8].

Staphylococcus aureus and *Escherichia coli* have frequently been isolated from patients with healthcare-associated infections, hospital-acquired invasive infections, bacteraemia and food borne infections

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[9–11]. Both *S. aureus* and *E. coli* are able to form biofilms and thus can form dual species community. *E. coli* and *staphylococcus* spp. infections are more virulent and/or result in worse outcomes than the single infections caused by either species [12].

Useful therapy for managing the antimicrobial resistance in poly microbial biofilms is still lacking. Hence, development of novel antimicrobial strategies against poly microbial biofilms infections is urgent need of time. Among the different antimicrobial strategies, non-invasive photodynamic inactivation or photodynamic antimicrobial chemotherapy (PACT) is effective against various bacterial infections [13]. PACT generates reactive oxygen species by a nontoxic light sensitive compound upon photo-illumination in presence of oxygen [14,15]. These reactive oxygen species oxidized cellular constituent and consequently drive cell towards its death [14,15]. Therefore, eradication of microorganisms by PACT is hypothesized to be a multi-target damaging process [16]. Therefore, the development of PACT-resistance by microorganisms is less than that of antibiotic resistance.

Vitamin B_2 or riboflavin is one of the vital micro-nutrients. It also found to be an excellent photosensitizer [17]. Photo-illuminated riboflavin shows a high ability to generate singlet oxygen [18]. Subsequently, these singlet oxygens cause oxidative stress mediated cell death in bacteria. On the other hand, riboflavin act as redox cofactors and in many key biological processes and has a safe potential to be used in humans [19]. Several reports have been showed that riboflavin used in PACT as a photosensitizer to eliminate various pathogenic microorganisms [20–26]. Most of the antibacterial studies of riboflavin used in PACT are against the planktonic form of non-multidrug resistant (MDR) strain of bacteria. Only few studies showed anti-biofilms activity of photo-illuminated riboflavin [27]. On the other hand, other photosensitizers (like methylene blue, toluidine blue etc.) were also reported to eradicate bacterial biofilm [28,29]. But the main advantage of riboflavin over other photosensitizers (like methylene blue, toluidine blue etc.) is that they are safe, natural substances without any reported side-effects [30]. Consequently, riboflavin can be considered as potential nontoxic photosensitizing agent in PACT particularly as antibiofilm agent. To the best of our knowledge, no study has been reported to describe its eradication potential against planktonic and sessile cells of mixed species of MDR bacteria. Moreover here in this study we used multi-antibiotic resistant *S. aureus* and *E. coli.* So, therefore, the aim of this study is to compare the effect of PACT using riboflavin on *S. aureus* and *E. coli* monoculture and as well as on dual culture.

2. Materials and methods

2.1. Bacterial strain

Clinically isolated culture of methicillin resistant *Staphylococcus aureus* (MRSA) and multi drug resistant *Escherichia coli* (MDR *E. coli*) are collected from Department of Medical Microbiology, Nil Ratan Sircar Medical College & Hospital, Kolkata, West Bengal, India and maintained in Department of Microbiology, Techno India University, West Bengal was used by growing the bacterial culture on Luria agar, and subcultured on Luria Broth prior to each experiment by adjusting turbidity to 0.5 O.D.

2.2. Photosensitizer and irradiation procedure

Riboflavin (RF, Sigma-Aldrich, St. Louis, MO) was used as a photosensitizer. Blue LED light of wavelength 450 nm used to expose the Photosensitizer. The Beam diameter was 4.5 cm and the ray height was 10 cm. Irradiation rate at that point was 40 W $m-^{2}$ as measured by a power meter (model: Laser Mate Coherent).

2.3. Determination of excited riboflavin and generation of superoxide radical in solution after photo-illumination

As a working stock 50 μM riboflavin was taken. UV spectra of riboflavin were recorded on UV–vis spectrophotometer (UV–vis Spectrophotometer 2206, SYSTRONICS) before and after photo-illumination. Generation of superoxide radical was determined after photo-illumination of riboflavin was determined by reduced nitro blue tetrazolium (NBT) method [31]. For this 50 μM riboflavin was added to the reaction mixture containing 33 μM NBT, 100 μM EDTA and 50 mM sodium phosphate buffer at pH 8.0 and 0.06 % Triton X 100 and the mixture was read at 560 nm.

2.4. Effect of PACT on the planktonic cell

PACT inactivation of bacterial cells by riboflavin was assessed on MDR *E. coli* and MRSA. 1:100 dilution of overnight culture of MDR *E. coli* and MRSA were taken in a fresh media separately and mixed into fresh media to form mixed culture. Then the mixed suspension was again incubated overnight at 37 °C after that secondary culture was prepared. 10^8 CFU/mL of separate culture as well as a mixed culture were treated with 50 μM of photo-illuminated RF, without photo-illuminated RF and only light. Control were left untreated, after treatment 10-fold serially dilution was prepared in PBS and then spread the treated and untreated cells on MacConkey agar (HiMedia) and Gelatin Mannitol Salt Agar (HiMedia) for isolation of *E. coli* and *S. aureus* respectively. Plates were kept at 37 °C for 24 h. Numbers of colonies were then counted. Only RF treated group without light was also evaluated.

2.5. ROS detection

Reactive oxygen species (ROS) detection was detected by using fluorescence spectroscopy using 2′,7′dichlorofluoresceindi acetate (DCFH-DA) [32]. At first overnight grown bacterial culture was centrifuged to collect the pellet. Pellet was then washed with phosphate buffered saline (PBS) and then finally re-suspended the pellet in PBS. 10^8 CFU/mL of both the bacterial suspensions was taken in eppendorfs individually as well as their mixed culture was incubated for 10 min with 2.5 μM (DCFH-DA) followed by the treatment of 50 μM of photo-illuminated RF, without photo-illuminated RF and only light. Control was left untreated. The fluorescence intensity was measured just after treatment at 525 nm with slit width 1.5 nm using spectro-fluorometer (Hitachi). Fluorescence intensity of control was recorded without treatment.

2.6. Measurement of lipid peroxidation

The amount of lipid peroxidation in bacterial suspensions was determined by assaying thiobarbituric acid-reactive substances formed [33]. 10 % SDS was added in treated cell samples and mixed thoroughly. After that, freshly prepared TBA was added to them and samples were incubated at 95 °C for 60 min.. Finally, samples were centrifuged at 5000 rpm for 15 min., the supernatant was collected to determine OD at 530 nm.

2.7. Effect of photo-illuminated riboflavin on respiratory chain lactate dehydrogenase activity

The lactate dehydrogenase (LDH) activity was measured by evaluating the reduction of NAD^+ to NADH and H^+ as lactate is oxidized to pyruvate [34]. The treated bacterial cell suspension was centrifuged and pellet was collected in a micro plate. To the collected pellet LDH reaction solution was added and the sample was incubated for 30 min at 25 °C. The OD was read at 490 nm.

2.8. Effect of PACT on Biofilms formation

Reduction of biofilms formation was investigated by the earlier reported method [35] with few modification. Briefly, overnight culture of MDR *E. coli* and MRSA were diluted to 10^8 CFU/mL into the fresh LB medium. 100 μl of the diluted bacterial suspension were inoculated in each well of 96 well micro titre plate the bacterial cell was treated with 50 μM of photo-illuminated RF, without photo-illuminated RF and only light. After treatment the bacterial cells were incubated for 48 h at 37 ◦C for biofilms formation. After incubation, the planktonic cells were removed and gentle washed the residual biofilms with PBS. Each biofilms wells were stained with 100 μL of 0.1 % crystal violet for 5 min on a shaker at room temperature. The excess dye was removed and 100 μL of 95 % ethanol was added in each well for the purpose of removing bound CV from a bacterial biofilm. Then optical density of the suspension was recorded by a micro plate reader at 630 nm.

2.9. Effect of PACT on mixed bacterial Biofilms

10^8 CFU/mL cells from overnight grown culture of MDR *E. coli* and MRSA were mixed together in fresh LB incubated for 24 h at 37 °C to form mixed culture. The mix bacterial culture broth was centrifuged at 10,000 rpm to pellet down the cells for 10 min and discard the supernatant. The cell pellet was then re-suspended in sterile PBS. Again 10^8 colony forming unit (CFU)/mL of mixed culture was taken and treated and allowed to biofilms formation as described above.

The effect of PACT on mixed biofilms was investigated by using two methods. Firstly, we quantified the biofilms formation by CV staining as described previously. Secondly, we visualized the biofilms formation of mix culture in presence and absence of PACT treatment by using light and confocal microscope.

For light microscopy, the biofilms assay was performed with small glass slides (1 × 1 cm) placed in the wells of the 24-well polystyrene plate. After treatment and subsequent incubation, planktonic cells were removed, and the biofilms formed on the glass slides were stained using crystal violet dye for 5 min. It was then gently washed with PBS and allowed to dry for 5 min. Then, the slides were viewed under a light microscope at 40x magnification and images were taken using digital camera.

For confocal microscopy, after treatment and subsequent incubation, glass slides were taken and washed with PBS followed by staining with acridine orange and observed using CLSM (Carl Zeiss LSM700, Jena, Germany)

2.10. Reduction of extracellular polymeric substance after PACT

Extracellular polymeric substance (EPS) production was estimated by using earlier reported method with few modification [36]. Briefly, treated (light, RF and RF + light) and untreated bacterial culture were incubated for 24 h at 37 °C. After incubation exhausted media were removed and biofilms were washed with PBS and 0.5 % NaCl was added. These cells suspended in 0.5 % NaCl were transferred to fresh sterile test tubes and added with equal volume of 5% phenol. To that solution, 5 vol of concentrated sulfuric acid containing 0.2 % hydrazine sulfate was added and incubated in dark for 1 h and the absorbance was measured at 490 nm.

2.11. Statistical analysis

Experimental values were expressed as mean ± S.E.M. of three independent experiments. All experiments were statistically analysed by one-way analysis of variance. $P < 0.05$ was considered statistically significant.

3. Results

3.1. Photo-illumination sensitized Riboflavin and generates superoxide radical

After photo-illumination we found the absorption maxima of riboflavin were decreased, but no change in absorption maxima was observed for RF without photo-illumination. Hence this was indicated that photo-illumination can sensitize riboflavin (RF). RF after light absorption undergoes intersystem conversion from singlet state to triplet state. During this intersystem conversion, photodegradation reaction can occur at both singlet and triplet state [37]. So, the reduction of absorption maximum of riboflavin after photo-illumination can be due to this photodegradation reaction [37]. Next, by reduced nitro blue tetrazolium (NBT) assay of RF, we found photo-illumination of RF was also significantly increased superoxide radical both dose and time dependently, as compared with the RF in absence of light (Fig.1A).

3.2. Photo-inactivation of planktonic bacteria

Using plate count method, we determined the effect of photo-illuminated RF on planktonic cells of MDR *E. coli*, MRSA and mixed culture. In this study, four study groups were used for MDR *E. coli*, MRSA and mix culture such as control (C) Light (L) Riboflavin (RF) Riboflavin + Light. After treatment monoculture of *E. coli* and *S. aureus* were plated on LA plates. Whereas, mixed culture was plated on MacConkey agar and Gelatin Mannitol Salt Agar for isolation of MDR *E. coli* and MRSA respectively. As shown in (Fig. 2A, B) reduction of approximately 5 and 4 log cfu/mL was observed after photo-inactivation of MDR *E. coli* and MRSA respectively. In mix culture (Fig. 2C, D) also we found planktonic cells of MDR *E. coli* and MRSA was reduced to 7 and 5 log cfu/mL respectively. Only RF treatment does not affect the viability of the planktonic bacterial cells but the group which are treated with light shows very little reduction of planktonic cells in *E. coli*, *S. aureus* and mix culture. We used other concentration of riboflavin to study its effect on planktonic cells in *E. coli, S aureus* and mix culture (Data not shown). As we found 50 μM concentration was the lowest concentration which after photo-illumination produced considerable singlet oxygen that was responsible for the significant antibacterial activity, we used this condition to study further.

3.3. Photo-illuminated RF increased intracellular ROS generation and oxidative stress

Cellular reactive oxygen species (ROS) levels can be measured in live cells by a technique that converts 2′,7′ –dichlorofluorescin diacetate (DCFDA). After enter into the cells, DCFDA is then deacetylated by cellular esterases to a non-fluorescent compound $DCFH_2$. In presence of ROS, this non fluorescence compound oxidized to a fluorescence compound 2′, 7′ –dichlorofluorescein (DCF) [38]. Therefore, the DCF fluorescence generated is directly proportional to the amount of ROS production in cells. MDR *E. coli*, MRSA and mix culture were exposed to photo-illuminated riboflavin shows significant increase of DCF Fluorescence (Fig. 3A). We found photo-illumination of riboflavin treatment increased DCF fluorescence intensity up to16.8, 14.8 and 16.3 folds in MDR *E. coli*, MRSA and mix culture respectively as compared with control. Whereas, control, light and RF group did not show any increase in fluorescence.

Next, we measured the oxidative stress induced by photo-illuminated RF in single and mixed bacterial culture. Lipid peroxidation is an important biomarker of oxidative stress. Malondialdehyde (MDA) is one of the final products of polyunsaturated fatty acids peroxidation in the cells. An increase in ROS causes overproduction of MDA. Thiobarbituric acid (TBA) assay is the most commonly used method for determination of the MDA. The assay is based on a condensation reaction of two molecules of TBA with one molecule of MDA, which is resulting in a

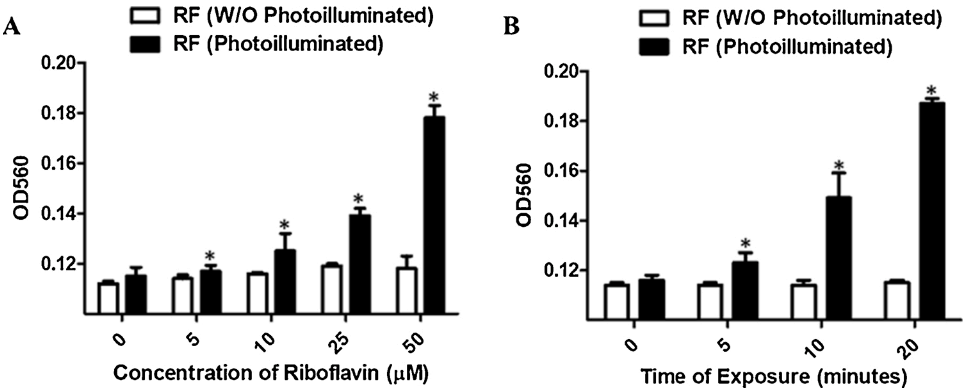

Fig. 1. A) Graphical representation of concentration dependent *in vitro* superoxide radical production by photo-illuminated and without photo-illuminated riboflavin. Values reported are mean ± S.E.M. of the three independent experiments. $*p < 0.05$ as compared to untreated cells **(B)** Graphical representation of time dependent *in vitro* superoxide radical production by photo-illuminated and without photo-illuminated riboflavin. Values reported are mean ± S.E.M. of the three independent experiments. $*p < 0.05$ as compared to.0 h.

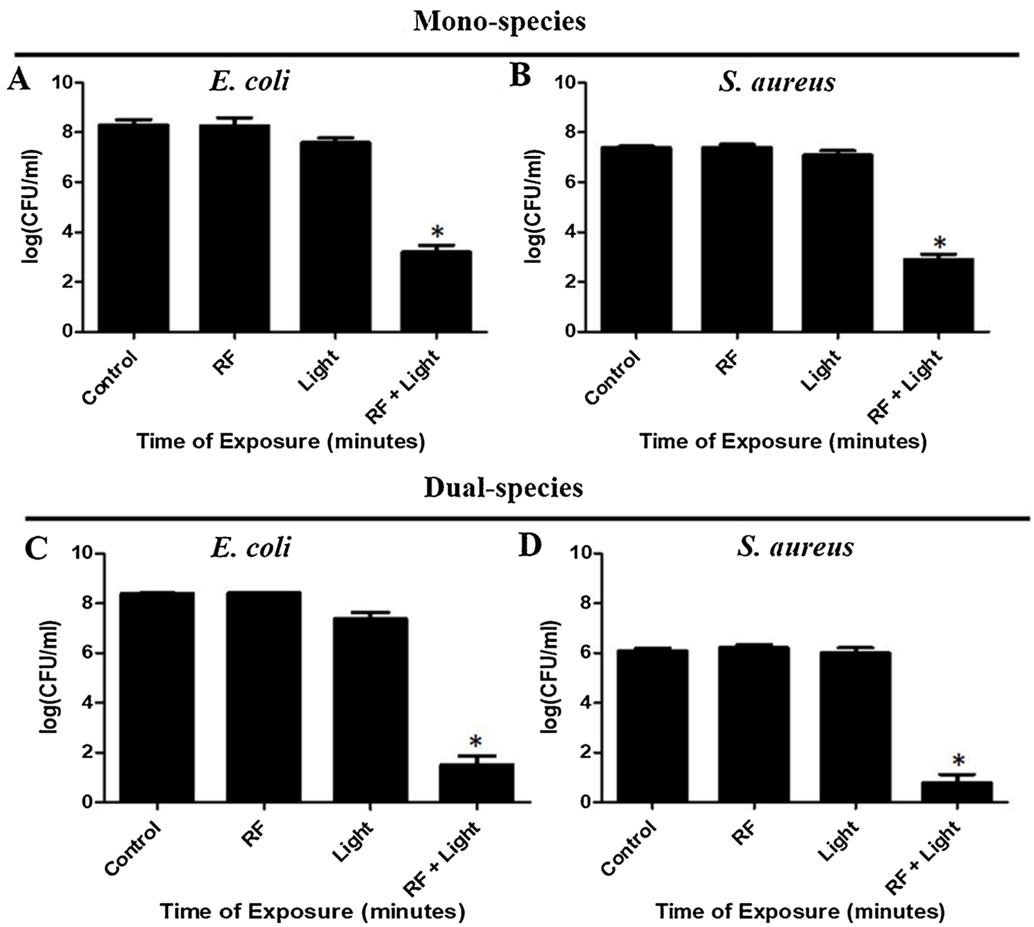

Fig. 2. Effect of photo-illuminated riboflavin on viability planktonic cells of **(A)** *E. coli*, **(B)** *S. aureus,* **(C)** *E. coli* in mix culture**(D)** *S. aureus* in mix culture. Values reported are mean ± S.E.M. of the three independent experiments. $*p < 0.05$ as compared to untreated cells.

colour compound TBA-MDA adduct, which can be determined spectrophotometrically [39]. We found in our study that, MDA level was significantly increased to 4.33, 3.8 and 3.92 folds in MDR *E. coli*, MRSA and mix culture respectively as compared with control (Fig. 3B). But MDA level in light and RF group are nearly similar to control.

3.4. Oxidative stress damaged respiratory system of the bacterial cells

During oxidative stress condition, mitochondrial enzyme lactate dehydrogenase (LDH) readily gets denatured. Therefore, LDH is considered as a biomarker to evaluate the ROS induced detrimental effect on the bacterial respiratory system [40]. In all three study groups such as MDR *E. coli*, MRSA and mix culture, LDH activity in cells treated with photo-illuminated riboflavin is considerably reduced as compared with control (Fig. 3C). However, LDH activity in light and RF group are nearly similar to control.

3.5. Reduction of biofilms formation by photo-illuminated RF

Initially, the inhibition of biofilms was estimated by crystal violet method. For both mono and mix culture, four groups have taken i.e. control, Light, RF, RF + Light. We found only Light and RF treatment shows no effect on biofilms forming ability of MDR *E. coli*, MRSA and Mix culture. However, as in (Fig. 4A, B and C) photo-illuminated RF

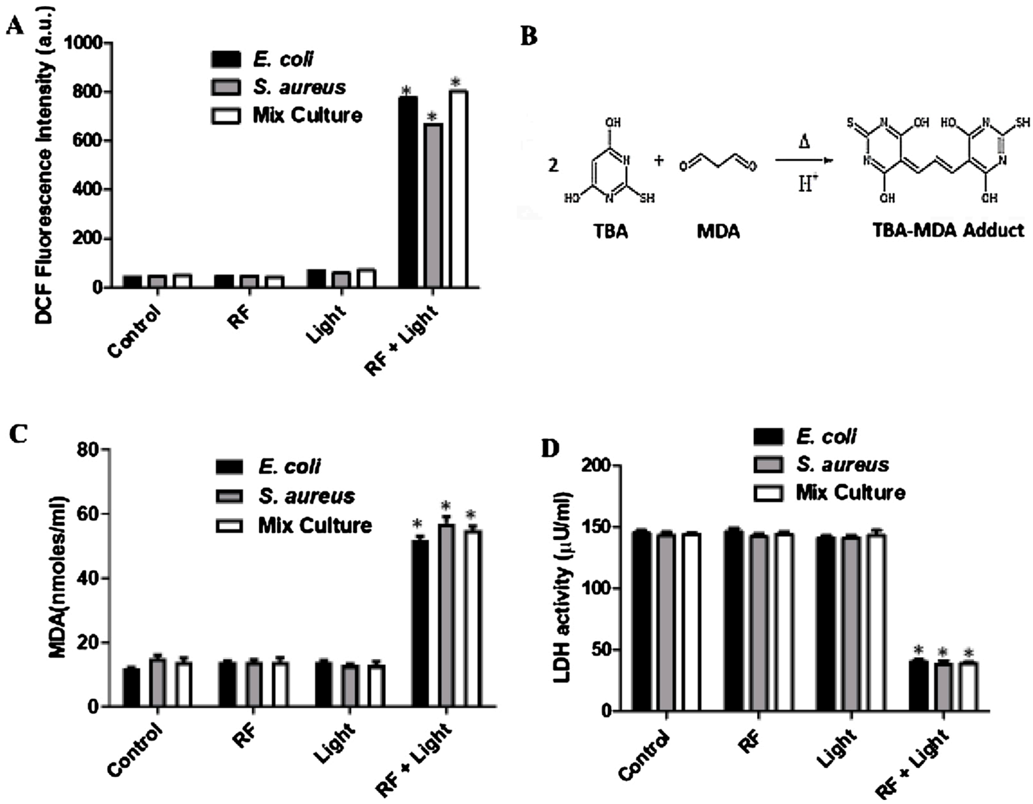

Fig. 3. **(A)** Detection of intracellular reactive oxygen species (ROS) by fluorescence intensity of DCF in *E. coli*, *S. aureus* and mix culture. Values reported are mean ± S.E.M. of the three independent experiments. *$p < 0.05$ as compared to untreated cells. **(B)** Chemical reaction of TBA-MDA adduct formation **(C)** Detection of lipid peroxidation by assaying TBA adducts formed in *E. coli*, *S. aureus* and mix culture. Values reported are mean ± S.E.M. of the three independent experiments. *$p < 0.05$ as compared to untreated cells. **(D)**Detection of the level of respiratory chain enzyme LDH in *E. coli*, *S. aureus* and mix culture. Values reported are mean ± S.E.M. of the three independent experiments..*$p < 0.05$ as compared to untreated cells.

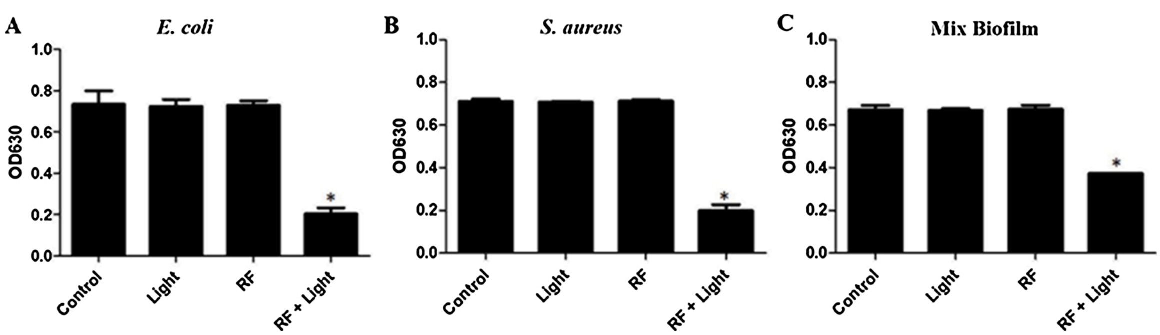

Fig. 4. Inhibition of biofilm formation: **(A)** *E. coli*, **(B)** *S. aureus*, **(C)** mix culture was treated with photo-illuminated and without photo-illuminated riboflavin. Values reported are mean ± S.E.M. of the three independent experiments. *$p < 0.05$ as compared to untreated cells.

inhibits biofilms formation by 34 % and 50 % of MDR *E. coli*, MRSA and Mix culture respectively as compared with control.

To investigate the mix culture biofilms inhibition by photo-illuminated RF we also used light and confocal microscopic study. Both the light and confocal microscopic study confirmed inhibition of mix culture biofilms by photo-illuminated RF. In light microscopic study, untreated samples retained more stains than the treated samples indicated disruption of biofilms (Fig. 5 A1-A2). Similarly, in CLSM, as compared with untreated control, decrease of acridine orange green fluorescence in photo-illuminated RF treated mix biofilms sample also indicating inhibition of biofilms formation (Fig. 5 A3-A4).

3.6. Reduction of EPS

For the formation and maintenance of biofilms EPS plays an important role. In our study EPS formation was estimated by phenol sulfuric acid methods. Four groups were taken i.e. control, Light, RF, RF + Light for MDR *E. coli*, MRSA and mix culture. EPS production was significantly reduced to and in photo-illuminated RF treated biofilms of MDR *E. coli*, MRSA and mix culture (Fig. 6A, B and C). In all groups only light and RF treatment shows no effect on EPS production.

4. Discussion

Recent emergence of antibiotic resistant and biofilms forming bacteria put a global threat since conventional antibiotic treatment becoming ineffective [41]. Biofilms provides its resident bacteria a safe environment and resistant to antimicrobial agents, toxic chemicals and host immune system [41]. For this reason, treatment of biofilms producing antibiotic resistant bacterial infection is very difficult [42]. *E. coli* and *S. aureus* are both opportunistic pathogens, biofilms producer and acquire different drug resistant gene. It has been reported that both of these bacteria form multispecies biofilms community. This type of multispecies biofilms community is much more resistant to antimicrobials than single species biofilms [12]. Therefore, alternative treatment strategies are required to combat against multidrug resistant bacteria within multispecies biofilms community.

Photodynamic Antimicrobial Chemotherapy (PACT) represents an

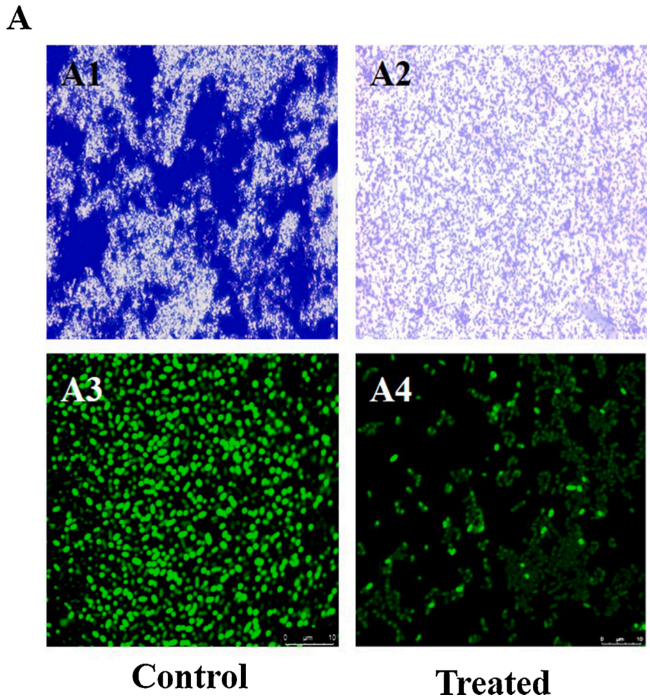

Fig. 5. A1-A2) Light Microscopic observation of photo-illuminated riboflavin induced mix species biofilm inhibition. **(A3-A4)** Confocal Microscopic observation of photo-illuminated riboflavin induced mix species biofilm inhibition.

attractive therapeutic strategy against drug-resistant microbes [43]. It can kill many bacteria within a few second without showing any resistant [44]. For an effective PACT an optimal photosensitizer is require, which should be biocompatible and non-toxic. Riboflavin (RF) is a nontoxic micronutrient but upon light exposure it undergoes intersystem conversion from singlet state to the triplet state. The triplet state reacts with available O_2 atom and generates oxygen radicals [17,45]. Only a few reports available which show inactivation of mix culture biofilms by using PACT. To the best of our knowledge, this is the first report, where we are providing the actual mechanism of inhibition of the mixed culture biofilms after PACT using riboflavin.

In our study we found, photo-illuminated riboflavin inhibits the survivability of both MDR *E. coli* and MRSA. The antibacterial effect of photo-illuminated RF is depending on extent of reactive oxygen species (ROS) production. In this study, we have estimated the ROS generation by using DCFH-DA. We found more ROS was generated in photo-illuminated RF treated *E. coli* than photo-illuminated RF treated MRSA. This is may be a reason why *E. coli* were little more susceptible to photo-illuminated riboflavin than *S. aureus*.

To determine the effect of Photo-illuminated RF on mix culture, we used two different media MacConkey agar and Gelatin Mannitol Salt Agar for *E. coli* and *S. aureus* respectively. We found photo-illuminated RF effectively reduced the no. of colony of both MDR *E. coli* and MRSA. But again *E. coli* are found more resistant to photo-illuminated RF than *S. aureus*. It may be due to in dual species biofilms *E. coli* was the out-competing species than *S. aureus* as reported earlier [46]. In our report also we found in mix culture *E. coli* shows much growth as compared to S. aureus. This indicates in mixed culture *E. coli* suppresses the growth of *S. aureus*.

Photo-illuminated RF inhibits mono-species biofilms formation of MDR *E. coli* and MRSA nearly same extent but we found mix species biofilms are much more resistant than mono species biofilms. This is similar to the previous reports suggested dual species biofilms are harder to eradicate than mono-species biofilms by antimicrobial treatment [47]. This is may be due to the interaction between different type of extracellular polymeric substance (EPS) production by these bacteria to form more viscous matrix as reported earlier [48]. This is also supported by our study as we observed reduction of EPS in dual species biofilms is less than the mono-species biofilms.

Extracellular polymeric substance (EPS) increased the antimicrobial resistant as it hinders the entry of antimicrobials to the interior of the biofilms. However, it has been reported earlier that, in PACT treatment, it is not important for the photosensitizer to penetrate inside the biofilms, because previously it has been reported that PACT treatment generates ROS inside the bacterial cells [49]. Generation of ROS may induce oxidative stress as we found level of lipid peroxidation increased significantly in photo-illuminated RF treated mono and mix culture bacterial cells, as similar to the previous report [50]. Finally, we found reduction of lactate dehydrogenase (LDH) activity in photo-illuminated RF treated cells indicating oxidative stress induced damaged of respiratory system leads to bacterial cell death.

5. Conclusion

We propose a photodynamic mechanism using riboflavin as a photosensitizer to target mono and dual species bacterial biofilm of MDR *E. coli* and MRSA. We concluded due to the interaction of more than one EPS produced in mix bacterial biofilms, it is more resistant than mono-species biofilms. Yet it is appeared that photo-illuminated riboflavin might be applicable for combatting disease caused by MDR *E. coli* and MRSA.

Declaration of Competing Interest

The authors declare that there is no conflict of interests regarding the publication of this paper.

Acknowledgment

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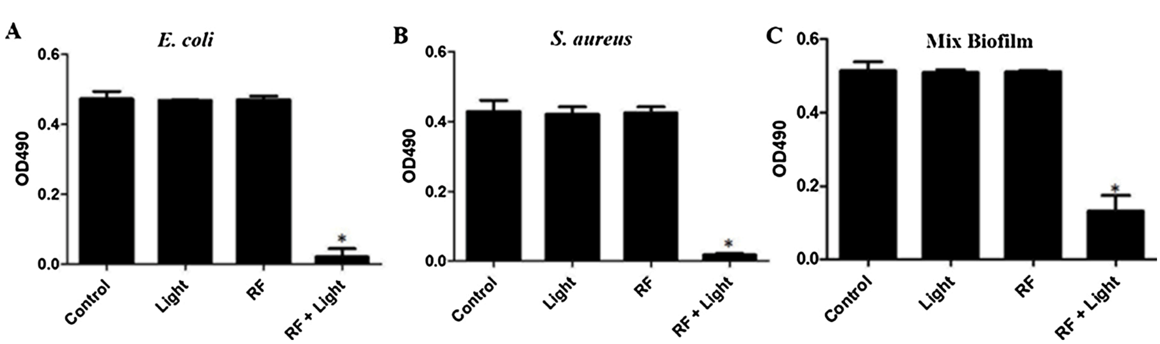

Fig. 6. Inhibition of EPS production: **(A)** *E. coli*, **(B)** *S. aureus*, **(C)** mix culture was treated with photo-illuminated and without photo-illuminated riboflavin. Values reported are mean ± S.E.M. of the three independent experiments. *$p < 0.05$ as compared to untreated cells.

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